

NTIRE 2026 Image Denoising ($\sigma = 50$) Challenge Factsheet -SwinIR-Based Transformer Network for Image Denoising-

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1. Introduction

Image denoising is a fundamental task in low-level computer vision that aims to recover clean images from noisy observations. Noise commonly appears during image acquisition due to sensor limitations, low-light environments, and electronic interference. Removing noise while preserving important image structures and textures is essential for many downstream vision applications such as image enhancement, object detection, and medical image analysis.

The NTIRE 2026 Image Denoising Challenge focuses on restoring images corrupted by additive Gaussian noise with a noise level of $\sigma = 50$. This setting is particularly challenging because high noise levels significantly degrade image details and structural information, making accurate reconstruction difficult.

Recent advances in deep learning have led to significant improvements in image restoration tasks. In particular, transformer-based architectures have demonstrated strong capability in modeling long-range dependencies and contextual information. In this work, we adopt a transformer-based denoising approach inspired by the SwinIR architecture and implemented using the BasicSR framework.

Our proposed method is designed to effectively remove strong Gaussian noise while preserving important structural details and textures in the reconstructed images. The model is trained using the NTIRE training dataset and evaluated using standard image quality metrics such as PSNR and SSIM.

2. Factsheet Information

2.1. Team details

- **Team name:** KLETech-CEVI
- **Team leader name:** Nikhil Akalwadi^{1,3}

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- **Rest of the team members:**

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- Sujatha C^{1,3}
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- **Affiliation of the team and/or team members with NTIRE 2026 sponsors (check the workshop website):**
NA

- **User names and entries on the NTIRE 2026 CodaBench competitions (development/validation and testing phases):**

CodaBench ID: awawq

- **Best scoring entries of the team during development/validation phase:**

PSNR: 29.95 dB

SSIM: 0.84

- **Link to the codes/executables of the solution(s):**

<https://github.com/CeviKle/NTIRE2026-KLETech-CEVI-Denoising>

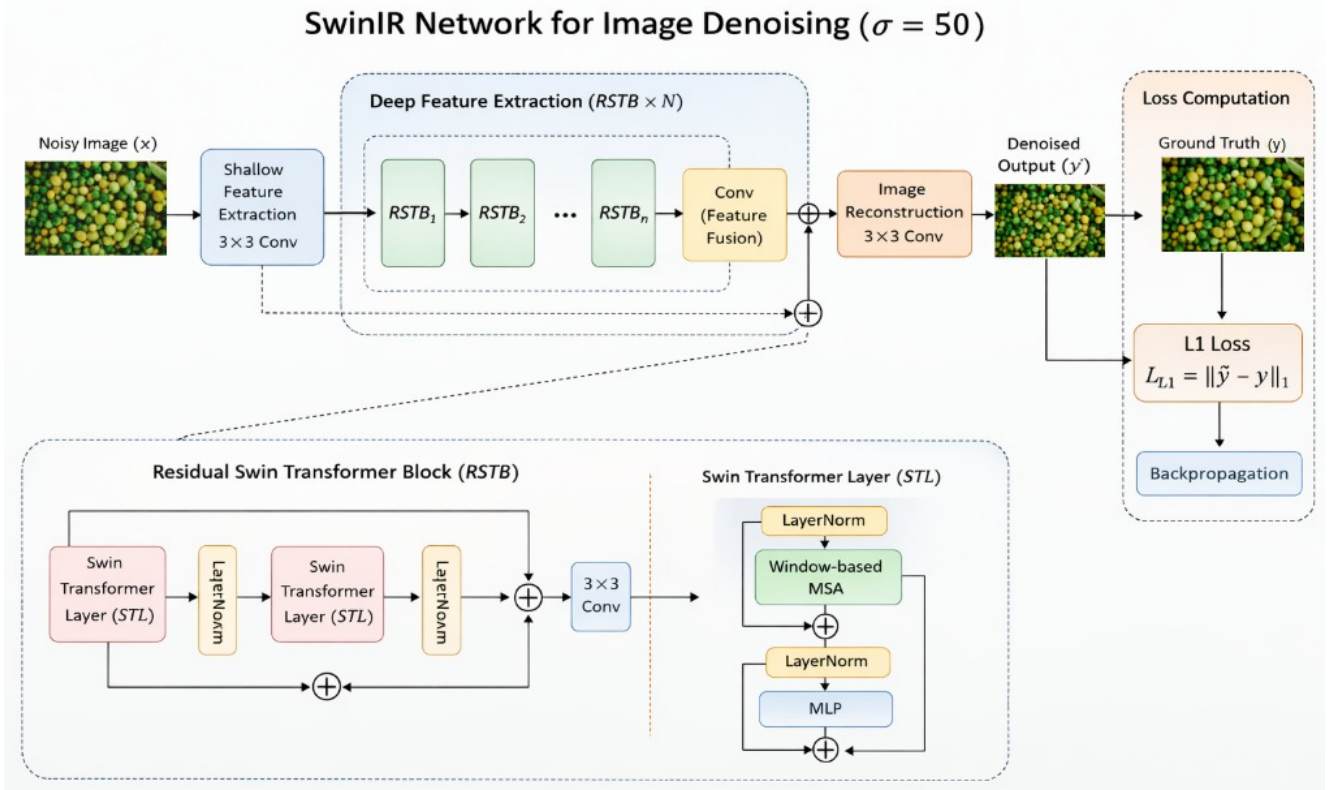


Figure 1. Overview of the proposed SwinIR-based image denoising framework. The network consists of shallow feature extraction, deep feature extraction using Residual Swin Transformer Blocks (RSTB), feature fusion, and image reconstruction. The model is trained using the L1 loss between the predicted denoised image and the ground truth clean image.

2.2. Method details

2.2.1. General Method Description

Our proposed solution is based on the SwinIR architecture [1] implemented within the BasicSR framework [2]. The overall architecture of the proposed model is illustrated in Figure 1. SwinIR is a transformer-based image restoration model that utilizes shifted window multi-head self-attention to capture both local and global contextual information efficiently. The network consists of three main stages: shallow feature extraction, deep feature extraction using Swin Transformer blocks, and image reconstruction.

Given an input noisy image x , the shallow feature extraction layer first maps the input image into a high-dimensional feature space using a convolution layer:

$$F_0 = H_{SF}(x) \quad (1)$$

where F_0 represents the shallow features and H_{SF} denotes the shallow feature extraction module.

The extracted features are then processed through multiple Swin Transformer blocks which capture long-range dependencies using window-based self-attention:

$$F_d = H_{STB}(F_0) \quad (2)$$

where F_d represents the deep features and H_{STB} denotes the stacked Swin Transformer blocks.

Finally, a reconstruction module generates the restored image $\hat{y}$:

$$\hat{y} = H_{rec}(F_d) \quad (3)$$

where H_{rec} denotes the reconstruction layer responsible for producing the final denoised image.

Figure 1 illustrates the overall architecture of the proposed SwinIR-based denoising framework. The model takes a noisy image as input, extracts shallow features using a convolution layer, and processes them through multiple RSTB for deep feature extraction. The extracted features are fused and passed through the reconstruction module to generate the final denoised output image.

2.2.2. Representative Pipeline

The overall denoising pipeline consists of the following stages:

1. Input noisy image with Gaussian noise level $\sigma = 50$

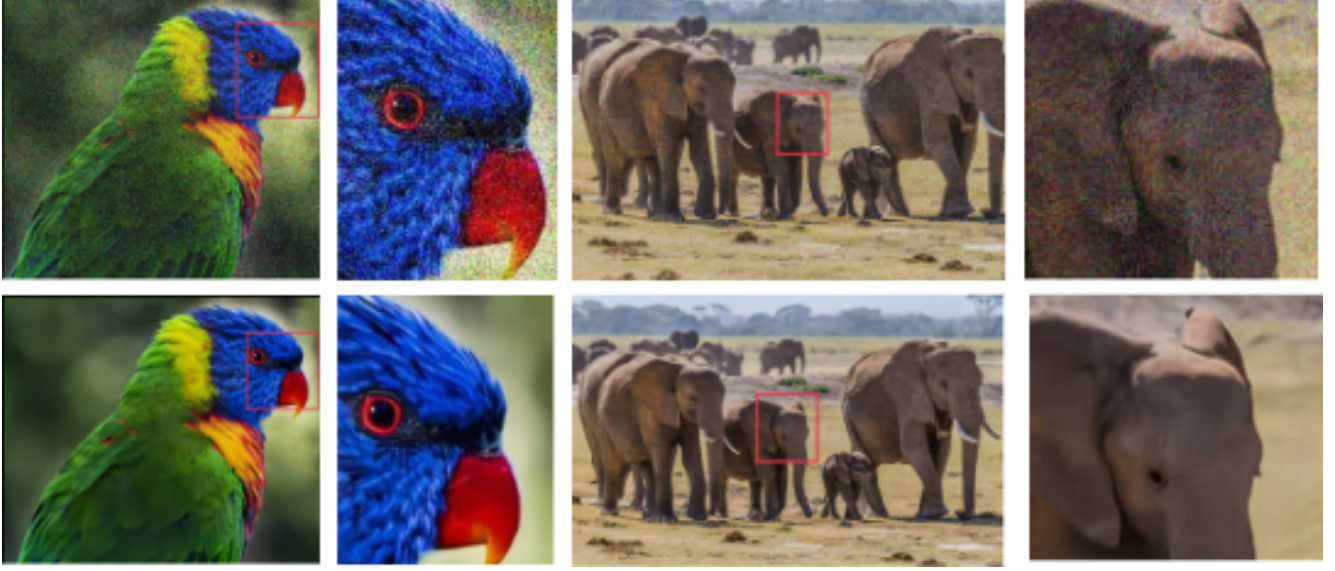


Figure 2. Qualitative denoising results of the proposed SwinIR-based method for Gaussian noise ($\sigma = 50$).

2. Shallow feature extraction using convolution layers
3. Deep feature extraction using stacked Swin Transformer blocks
4. Reconstruction module to generate the denoised image

2.2.3. Training Strategy

The model is trained using the dataset provided in the NTIRE 2026 Image Denoising Challenge. During training, images are randomly cropped into patches of size 128×128 to increase the number of training samples. Data augmentation techniques such as horizontal flipping and rotation are applied to improve generalization.

The network is optimized using the Adam optimizer with an initial learning rate of 2×10^{-4} and weight decay of 1×10^{-4} . A MultiStep learning rate scheduler is used to gradually reduce the learning rate during training. The model is trained for 100,000 iterations with a batch size of 4 on single NVIDIA RTX3090.

The training objective is the L1 pixel loss, which minimizes the difference between the predicted denoised image $\hat{y}$ and the ground truth clean image y :

$$L_{L1} = \|\hat{y} - y\|_1 \quad (4)$$

2.2.4. Method Complexity

The proposed model uses an embedding dimension of 96 with four transformer stages, each containing six Swin Transformer blocks. Each stage utilizes six attention heads and a window size of 8. The architecture provides a good balance between restoration performance and computational efficiency.

2.2.5. External Models and Additional Data

No external datasets or pre-trained models were used during training. The model was trained solely using the dataset provided by the NTIRE 2026 challenge.

2.2.6. Training Description

Training was conducted using the PyTorch framework with the BasicSR toolbox. The model was trained for 100,000 iterations using the Adam optimizer and the L1 loss function. Data augmentation techniques including horizontal flipping and rotation were applied during training.

2.2.7. Testing Description

During testing, the trained model checkpoint is loaded and applied to the noisy images from the NTIRE test dataset. Each image is processed individually, and the network outputs the corresponding denoised image.

2.2.8. Advantages of the Proposed Solution

The transformer-based design enables the model to capture long-range dependencies and contextual relationships between image regions. This improves the ability of the network to remove strong Gaussian noise while preserving important textures and structural details in the restored images.

Figure 2 shows qualitative examples of the denoising results obtained using the proposed method. The first row shows the noisy input images, while the second row shows the corresponding denoised outputs generated by our model. The results demonstrate that the proposed approach effectively removes noise while preserving fine image details and structural information.

3. Other details

- **Planned submission of a solution(s) description paper at NTIRE 2026 workshop:**

Yes

- **Other comments: encountered difficulties, fairness of the challenge, proposed subcategories, proposed evaluation method(s), etc:**

The NTIRE 2026 Image Denoising Challenge provided a valuable platform for evaluating modern deep learning-based image restoration methods. The dataset and evaluation protocol enabled fair benchmarking under high noise conditions. Although some minor difficulties were encountered during model submission and evaluation, the overall challenge organization was well structured and provided an effective environment for developing and testing image denoising models.

References

- [1] Jingyun Liang, Jiezhang Cao, Guolei Sun, Kai Zhang, Luc Van Gool, and Radu Timofte. Swinir: Image restoration using swin transformer. In *ICCV Workshops*, 2021. 2
- [2] Xintao Wang and Kai Zhang. Basicsr: Open source image restoration toolbox. *arXiv preprint arXiv:2012.02181*, 2021. 2